RESULTS SUMMARY - Lockheed Martian

THE DRAKKAR ASCENT CHALLENGE

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1. Design Basis

Parameter	Value
Shell length	1500 mm
Applied load factor (team-selected)	**1.1** - multiplies F, M, V and p before analysis
Required safety factor	1.1
Moment case	Brief (primary moment only)
Pressure path	Wall-only strength (no closed-end credit)
Diameter search range	1 - 300 mm
Thickness step	0.01 mm

Applied loads (after x1.1 factor)

Symbol	Value
Axial thrust P	33 000 N
Shear V	1 611.6 N
Internal pressure p	330 000 Pa
Bending moment M	3 027.9 N·m (first-order: 2 417.5 N·m)

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2. Selected Metal Design

2a. Primary moment case (bending only)

Parameter	Value
Material	Aluminium 7075-T6
Outside diameter	**119 mm**
Wall thickness	**0.96 mm**
Shell mass	**1.501 kg**
Governing failure mode	Shell interaction (local)
Governing utilisation U	0.9962
Strength utilisation	0.7830
Euler (column) utilisation	0.1692
Ovalisation utilisation	0.7425
r/t ratio	61.5
L/r ratio	25.4

2b. Separate M + V·L case (transfer loads)

Parameter	Value
Material	Aluminium 7075-T6
Outside diameter	**137 mm**
Wall thickness	**1.10 mm**
Shell mass	**1.980 kg**
Governing failure mode	Shell interaction (local)
Governing utilisation U	0.9956
Strength utilisation	0.8486
Euler (column) utilisation	0.0968
Ovalisation utilisation	0.8884
r/t ratio	61.8
L/r ratio	22.1

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3. Full Material Trade - Primary Moment Case (LF = 1.1)

Optimal shell for each material at the team load factor. All candidates governed by shell interaction.

Material	D (mm)	t (mm)	Mass (kg)	U
Aluminium 7075-T6 ✓	**119**	**0.96**	**1.501**	0.9962
Magnesium AZ31B	169	1.10	1.540	0.9978
Aramid/Epoxy QI	169	1.23	1.361	0.9951
Carbon/Epoxy QI	101	1.20	0.903	0.9936
Titanium Ti-6Al-4V	96	0.84	1.669	0.9978
Glass/Epoxy QI	125	1.60	1.768	0.9955
Steel 4130	154	0.58	3.292	0.9999

> **Carbon/Epoxy QI** achieves the minimum shell mass (0.903 kg) but was not selected as the primary metal candidate. Aluminium 7075-T6 is the primary metal selection at 1.501 kg.

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4. Full Material Trade - Transfer Load Case (Separate M + V·L)

Material	D (mm)	t (mm)	Mass (kg)	U
Aluminium 7075-T6 ✓	**137**	**1.10**	**1.980**	0.9956
Magnesium AZ31B	199	1.23	2.029	0.9987
Carbon/Epoxy QI	113	1.40	1.178	0.9946
Aramid/Epoxy QI	195	1.40	1.788	0.9984
Titanium Ti-6Al-4V	111	0.96	2.205	0.9915
Glass/Epoxy QI	141	1.84	2.293	0.9988
Steel 4130	183	0.65	4.385	0.9979

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5. Sensitivity to Load Factor

Minimum-mass Aluminium 7075-T6 shell at five load factor levels:

Load factor	D (mm)	t (mm)	Mass (kg)
1.00	115	0.93	1.405
1.10	**119**	**0.96**	**1.501** ← selected
1.25	120	1.04	1.638
1.50	130	1.09	1.861
2.00	144	1.21	2.288

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6. Pressure Scenario Utilisations

Pressure-only strength utilisation (no combined loads) for each material at the selected geometry:

Material	Wall-only U	Closed-end U
Aluminium 7075-T6	0.783	0.763
Titanium Ti-6Al-4V	0.799	0.788
Steel 4130	0.829	0.782
Magnesium AZ31B	0.826	0.770
Carbon/Epoxy QI	0.804	0.793
Glass/Epoxy QI	0.787	0.766
Aramid/Epoxy QI	0.764	0.709

All candidates pass wall-only pressure strength with utilisation < 1.0.

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7. Assumptions and Scope

- Uniform nominal wall; pinned beam-column envelope.
- No pressure-stabilisation credit applied.
- Equivalent-QI quadratic failure model for composites.
- Mass includes the shell only (no end rings, fittings, or connections).
- Prescribed end rings retain circularity; shell stresses carry bending.
- Local interaction criterion governs all selected candidates.
- Transverse-shear interaction handled via surrogate; long-shell correlation extrapolated.
- Material strength cap active for Aluminium 7075-T6 (σ_bending_cr capped at yield).

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8. Verification and Interpretation

- All **16 unit tests pass**. Independent arithmetic reproduces 14 detailed candidates and 35 load-scenario checks.
- Results satisfy the implemented analytical screen.
- **Model limitations:** inelastic capacity, transverse-shear interaction, and long-shell applicability not fully validated; a total-assembly optimum is not established.
- Full verification data: `results/calculation_verification.json`
- Raw trade sweeps: `results/trade_study.csv`, `results/transfer_trade_study.csv`

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9. Output Files

File	Description
`output/pdf/Drakkar_Analytical_Report.pdf`	Full analytical report with equations and worked calculations
`output/pdf/Drakkar_Results_Summary.pdf`	PDF version of this summary
`assets/trade_study_diameter_vs_mass.png`	Shell mass vs. diameter sweep (all materials)
`assets/failure_modes_vs_diameter.png`	Failure mode utilisations vs. diameter
`assets/trade_materials_comparison_barchart.png`	Bar chart comparing all-material minima
`assets/thrust_barrel_cad_drawing.png`	Engineering drawing of thrust barrel
`results/design.json`	Full numerical results (JSON)
`results/trade_study.csv`	Diameter sweep data - primary case
`results/transfer_trade_study.csv`	Diameter sweep data - transfer case

Updated: 2026-09-30. All data sourced directly from `results/design.json` produced by `parametric_diameter_study.py`.